Monocular 4D Human–Object Reconstruction: a modular pipeline, physical refinement and failure analysis

V2D Challenge 2026, Track 1 (Reconstruction) — team AI_CENTER — technical report, draft of 1 November 2026

Executive summary

We reconstruct, from a single static third-person video, the MHR body of the actor and the 6-DoF pose, mesh and scale of the manipulated object for all 30 Track 1 episodes (10 object types, 9,877 scored frames). The system chains pretrained models (AnyCalib, MoGe, Detectron2, SAM3D-Body/MHR, Grounding DINO, SAM2, SAM-3D object generation, FoundationPose) with our own fitting and refinement code. The final submission, **full_v2**, was produced by running the committed code on all 30 videos in a fresh directory (stages 1–8 followed by one post-processing script); the run had to be resumed after four pipeline bugs were fixed (section 7).

Metric (lower is better)	Control (27 Oct)[1]	Best earlier (phys_v7)	Final: full_v2	Change vs control	Leaders (≈, 27 Oct)
CD-H (cm)	15.97370	15.85531	15.87878	–0.6 %	12–13
CD-O (cm)	30.95619	27.91838	**27.75448**	–10.3 %	16–18
ACC-H (cm/frame²)	0.17421	0.14129	**0.14109**	–19.0 %	≈0.135
ACC-O (cm/frame²)	0.11673	0.08385	**0.08147**	–30.2 %	≈0.069
PEN (cm)	0.04496	0.00921	**0.00826**	–81.6 %	≈0 (CARI4D baseline 0.00423)

[1] Control = *stool_static_plus_smooth_t150*, the last submission before this work package with the same scorer version. All Kaggle numbers in this report are leaderboard values reported by the team; upload receipts were not available on the compute server.

Main findings

- **Temporal regularisation** gave the most reliable gains: stronger object smoothing ($\lambda = 1000$) and representation-safe body smoothing ($\sigma = 4$) cut ACC-O by 30 % and ACC-H by 19 %, and both were predicted correctly by a local test set before submission.
- **Penetration** fell from 0.045 to 0.008 cm. Pushing objects off the hands works but costs CD-O; moving the hands instead (upper-limb refinement with image and motion constraints) kept CD-O and was only accepted after human review rejected two visually wrong variants.
- **The largest single CD-O gain (–2.87 cm) came from fixing one grossly misplaced object** (a hula hoop lying on the floor around the feet that stage 8 had placed 50–70 cm in front of the person), together with an image-ellipse fit for a held hoop and two static-desk locks. CD-O is a mean over 30 episodes, so single gross errors dominate it.
- **Depth corrections learned off-benchmark failed twice** (CD-O +5.9 and +6.0 cm). A silhouette-based object refinement improved image overlap but did not move objects towards the reference; it is disabled.
- **Remaining gap:** CD-O (27.8 vs 16–18 cm) is dominated by object placement relative to the body, which neither better meshes nor image-fit refinement corrected in our tests. CD-H was barely touched.

Data-use note. A local test set built from Track 2 Tier 1 (multi-view reference poses and meshes) was used to *evaluate* pipeline variants and choose two hyperparameters (λ and body σ). No Track 2 pose, mesh or video was used as an input to any Track 1 reconstruction. Whether evaluation-only use is permitted must be confirmed against the challenge rules before this report is published (section 5.3).

1. Task and evaluation

Input. One RGB video per episode (1536×1152, 30 fps, static exocentric camera), the object name and a text prompt. 30 episodes: hula hoop, big red bowl, iron, white desk, black pan, foam grass block, paint roller, pink foam roll, short wood stool, white laptop cart (three videos each). Only frames listed in the public sample (9,877 "scored" frames, often several stretches per episode) are evaluated.

Output per episode. MHR body parameters (pose [T,136], scales [68], shape [45]); object rotation and translation per frame; one object scale; one object triangle mesh (≤ 4,096 vertices/faces after the kit's budgeting). Everything is in the camera frame, in metres.

Scoring (read from the official metric code, `metric_code/track_1`):

- One similarity transform (Sim(3)) is fitted per episode between our MHR vertices and the reference on the *first scored frame only*, and applied to every frame. Any later drift of the body, and any error of the object relative to the body, is therefore charged in full.
- CD-H / CD-O: symmetric Chamfer distance (sum of the two directed mean nearest-neighbour distances) between sampled human / object surfaces, per frame, in cm.
- ACC-H / ACC-O: error of second frame differences of 22 body joints / object translation, only over consecutive scored frames.
- PEN: mean depth of 512 fixed hand vertices inside our own object mesh (winding-number inside test), converted with the alignment scale. It needs no reference, so it can be computed locally; for every submission the Kaggle value was 81–93 % of our local estimate.
- Each metric is first averaged over an episode's frames, then over the 30 episodes (episodes weigh equally).

Compute. Two NVIDIA A40 GPUs (devices 0 and 2; device 1 belonged to another user). One environment per model family under /opt/venvs. A full run of stages 1–8 on all 30 videos takes about 19 hours on two GPUs; post-processing about 1–1.5 hours.

2. System architecture

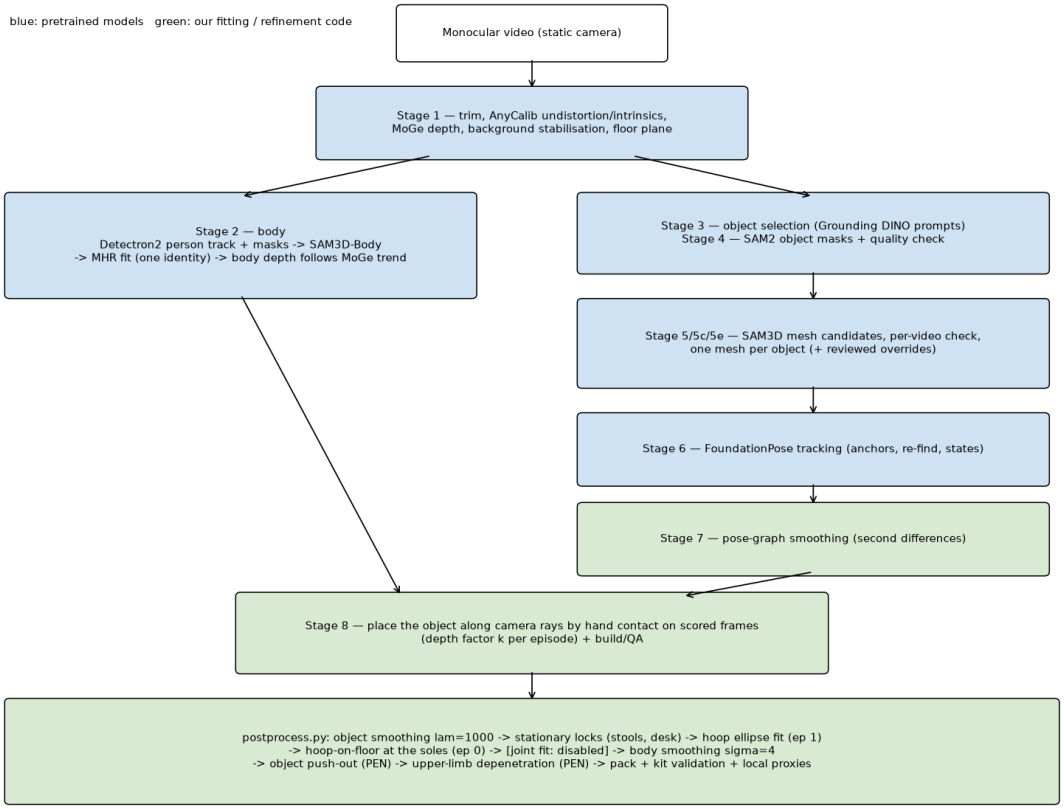


Figure 1. The pipeline as it produced full_v2. Blue: pretrained models; green: our fitting and refinement code. Stage outputs are cached per episode with "stamps" (hashes of code, settings and inputs) so a stage re-runs exactly when something it depends on changes.

Stage	What it does	Key outputs
1 prep	Trim to the scored span ±10 frames; AnyCalib estimates intrinsics/distortion from 16 frames per video (one camera per video); MoGe per-frame depth stabilised against the static background; RANSAC floor plane.	frames, depth, intrinsics_stable.json, floor.json
2 body	Actor track and person masks (Detectron2); SAM3D-Body per frame; one-identity MHR fit with temporal smoothing; the body's distance is nudged to follow MoGe's slow depth trend.	body/mhr_fit.npz
3–4 masks	Grounding-DINO prompts select the object; SAM2 propagates masks, with re-detection on lost stretches, leak repair and a per-frame quality assessment.	masks/, masks_quality.json
5 / 5c / 5e mesh	SAM-3D object meshes from several frames; per-video silhouette check; one mesh per object chosen over its three videos and validated on held-out frames. Reviewed overrides exist for the iron and hula hoop.	mesh_scaled.obj, mesh/validate.json
6 track	FoundationPose registration at quality-ranked anchor frames, forward/backward tracking, re-finding and per-frame states (observed / tracked / occluded / lost).	poses/, track/states.json
7 smooth	One sparse least-squares solve per episode: stay close to the tracker (weighted by IoU, outliers and hidden frames down-weighted) while penalising second differences.	poses_smoothed/
8 place	Scale the object along the camera rays by a factor k (image projection unchanged) chosen from hand contact and penetration on scored frames; build the episode NPZ; QA; render review frames.	submission/episodes/*.npz
post	track1/postprocess.py with postprocess_config.json (one entry per step, each with the reason it exists); see section 3.	experiments/<run>/track1.parquet

3. Refinement methods

All refinements run after stage 8 on the packed per-episode arrays, each in its own experiment directory, and each declares which arrays it may change; a merge step refuses changed meshes or array layouts.

3.1 Object smoothing (stage 7, $\lambda = 1000$)

Per translation axis the smoothed trajectory x minimises $\Sigma_f\, w_f(x_f - raw_f)^2 + \lambda\, \Sigma_f(x_{f-1} - 2x_f + x_{f+1})^2$, where $w_f = \mathrm{clip}((IoU_f - 0.3)/0.4, 0.05, 1)$. Rotations are treated the same way as small rotation vectors about a slowly varying reference. λ was 50 in the original pipeline. On the local test set (section 5.3) $\lambda = 1000$ gave the best CD-O and an ACC-O equal to that of a perfect constant-velocity guess (Figure 2). Stage 8's depth placement is pinned to production while re-smoothing, so smoothing and placement can be judged separately. Episode 12 (fast back-and-forth pan wiping) keeps $\lambda = 50$ because $\lambda = 1000$ moved well-fitted frames by 23 cm (p95).

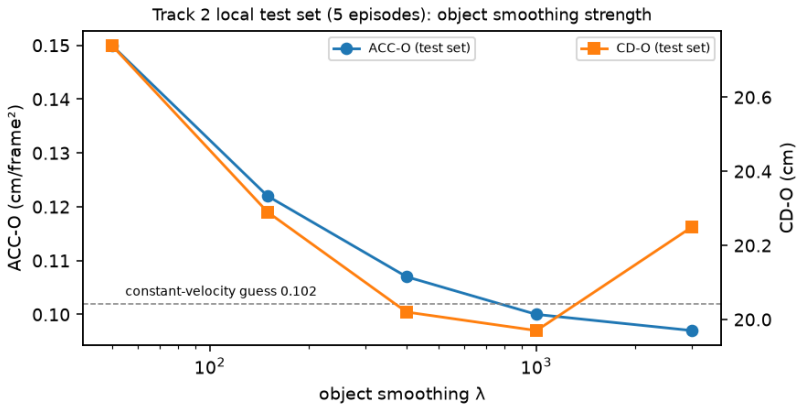


Figure 2. Object smoothing strength on the local test set (5 scoreable episodes). Kaggle confirmed the choice: λ 150 → 1000 lowered ACC-O by 0.021 and CD-O by 0.18 (phys_v4).

3.2 Body smoothing ($\sigma = 4$)

MHR joint rotations are Euler angles, and the per-frame fit sometimes jumps to an equivalent angle set (2π wraps of the global rotation; the alternative Euler triple for a joint). Gaussian smoothing across such a jump averages two encodings of one rotation into a wrong pose. `smooth_body.py` unwraps the global rotation, smooths root translation everywhere, smooths rotations between jumps, and tapers back to the raw fit near each jump. Two simpler variants failed on the test set (pelvis acceleration error 0.149 → 0.172; joint self-acceleration 0.13 → 0.27) before this one reduced the pelvis error to 0.122. On Kaggle it improved ACC-H 0.174 → 0.141 and, unexpectedly, CD-H by 0.12 cm.

3.3 Penetration (PEN)

Two complementary steps, run after smoothing:

- **Object push-out** (`pen_squeeze.py`): per scored frame, the bounded translation that moves the object off the hand points (scorer-exact inside test on the budgeted mesh), smoothed within each scored stretch, capped at 5 cm. Where both hands squeeze the object from opposite sides (foam block) no translation helps; the object may then shrink by up to 10 %, choosing the least shrink within 0.01 cm of the best held-out PEN. Each episode is accepted only if PEN drops on a held-out set of hand points and object jitter does not rise. Cost: CD-O rises with the push distance (phys_v5: PEN –45 %, CD-O +0.11).
- **Upper-limb depenetration** (`hand_depen.py`): the objects stay byte-identical; finger, wrist, elbow and shoulder parameters are optimised against a signed-distance grid of the packed mesh. Losses: hand points outside the surface; the closest 5 % of hand points may not move away from the surface (contact kept); hand vertices keep their image projection (≤ 3 px free); 3D motion beyond 2 cm is penalised (a soft cap); parameter changes are regularised and smoothed over time.

The final configuration was reached through visual review (Figure 3): the first version removed penetration by bending fingers out of the object; the second kept the hands right in the image but hid one hand 26 cm behind a desk, visible only in a top-down view; the third doubled body jitter until corrections were smoothed. On Kaggle the accepted version lowered PEN 0.0150 → 0.0091 while CD-O improved by 0.085 (phys_v6) — although the object arrays were identical, because CD-O depends on the body through the first-frame alignment.



Figure 3a. Rejected: finger-only depenetration (foam block, episode 16). Red: original hand, green: corrected hand, orange: object. PEN fell to ~0, but the corrected fingers point out of the block.

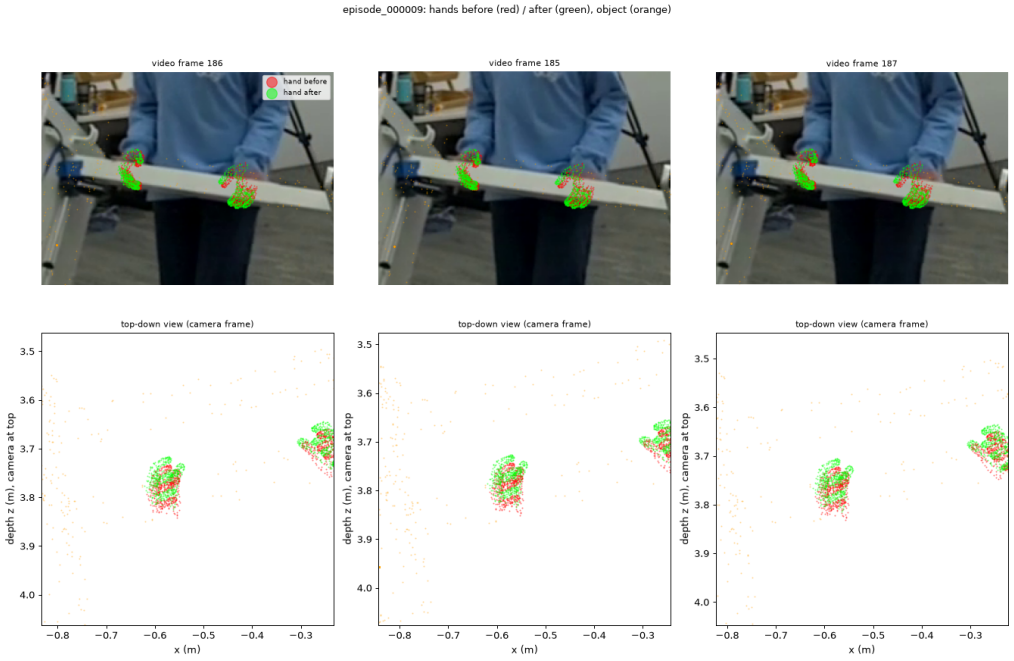


Figure 3b. Accepted variant (desk, episode 9). Top: camera view; bottom: top-down view in the camera frame, which exposes depth changes the camera view cannot show.

3.4 Stationary objects

For objects the team confirmed to be static over a known interval (stools in episodes 24 and 26 for the whole video; the desk in episode 10 until it is moved and in episode 11 until the end), one pose replaces the per-frame poses. It is chosen among clear anchor frames spread over the interval by how well it explains the masks of the other anchors, and accepted only if held-out frames lose ≤ 0.05 mask IoU and nothing moves more than 8 cm. The end of a still interval is detected from the raw tracker (> 4 cm for 10 frames). In the fresh run the detector found a shorter still interval for episode 11 (to frame ~177 instead of ~704); the gate accepted it, so the result is safe but smaller.

3.5 Hula hoop: geometry instead of tracking

A thin, nearly symmetric ring is the worst case for both mask-based tracking and generated meshes. Before this work package, three stage-6 interventions (mask-hole repair, visibility- and quality-ranked anchors) were rejected by scored-frame gates. Two geometric methods succeeded:

- **Held hoop (episode 1),** `hoop_conic.py`**.** SAM2 segments the ring cleanly even when FoundationPose loses it. A circle of known radius seen by a calibrated camera is determined by its image ellipse up to a two-fold tilt; we fit the ellipse, solve both candidates (perspective-exact least squares), pick the one whose rim passes closest to a hand, and keep it per frame only where it explains the mask better than the tracker's pose;

a final λ = 1000 solve removes switching jumps. Synthetic tests recover circles to 3 cm / 3°.

- **Hoop lying on the floor (episode 0),** `hoop_floor.py`**.** The video shows the person standing inside a hoop on the floor; stage 8 had placed it 22–29 cm above the soles and 50–70 cm in front of the body, because it placed objects by hand contact while the contact was with the feet. On frames where the ring is flat, the factor that brings its lowest point to the sole level is computed per frame; if ≥ 30 frames agree within 5 %, the object is scaled about the camera by their median (×1.193; ×1.197 in the fresh run). Feet inside the ring: 0 % → 56 %.

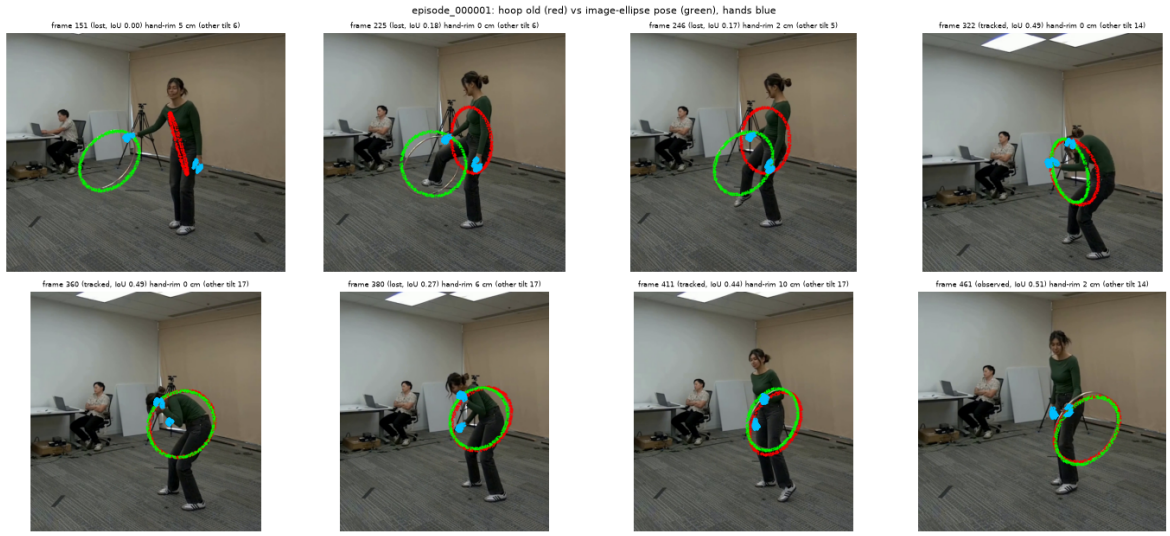


Figure 4a. Episode 1: tracked hoop (red) vs image-ellipse pose (green), hands blue. On frames where the tracker was lost the hoop moves from the body to the hand that holds it.

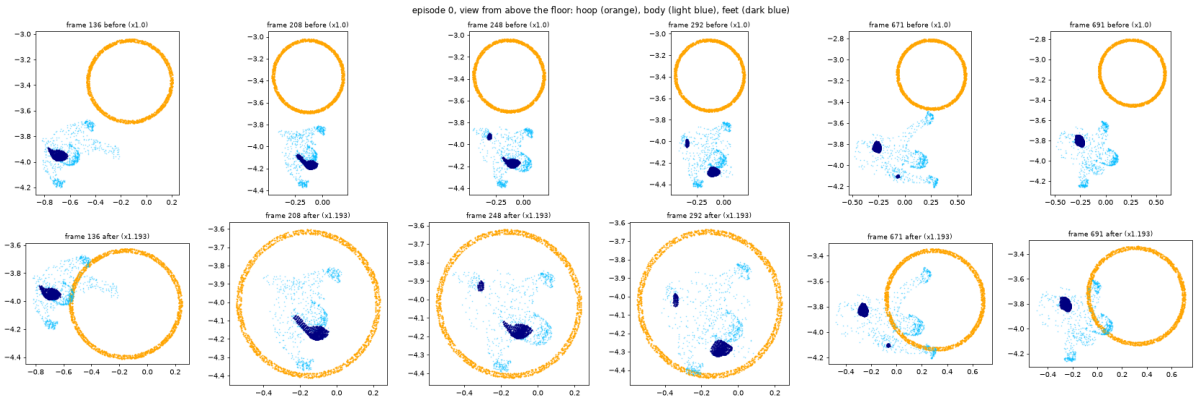


Figure 4b. Episode 0 seen from above the floor plane: hoop (orange), body (light blue), feet (dark blue), before (top) and after (bottom) the floor correction.

These changes (with the two desk locks) lowered Kaggle CD-O from 30.79 to 27.92 cm. Because they were submitted together, the share of each is not measured; the size of the gain (≈86 cm on one episode) and the magnitude of the episode-0 error make it likely that most of it comes from episode 0.

3.6 Joint object refinement (built, tested, disabled)

`joint_fit.py` refines all frames of an episode at once: per-frame object rotation and translation (body and object scale fixed) against SAM2 masks via differentiable silhouettes (nvdiffrast), optional hand contact and no-penetration terms on an SDF of the mesh, second-difference smoothing, and a prior that resists depth changes more than lateral ones. It is *not* the full body–object–hand optimisation discussed during the work (no body variables, no depth term). Results: with hand terms the object shook (self-acceleration 0.025 → 0.286) or moved 4.8 cm in depth; silhouette-only improved mask IoU (+0.026 to +0.04 on three videos) with shifts ≤ 2.5 cm, but on a test-set video the error against the reference did not change (CD-O 23.5 → 23.5 cm, lateral error 12.1 → 12.2 cm). It is therefore disabled. This rules out this implementation, not joint optimisation in general.

4. Results

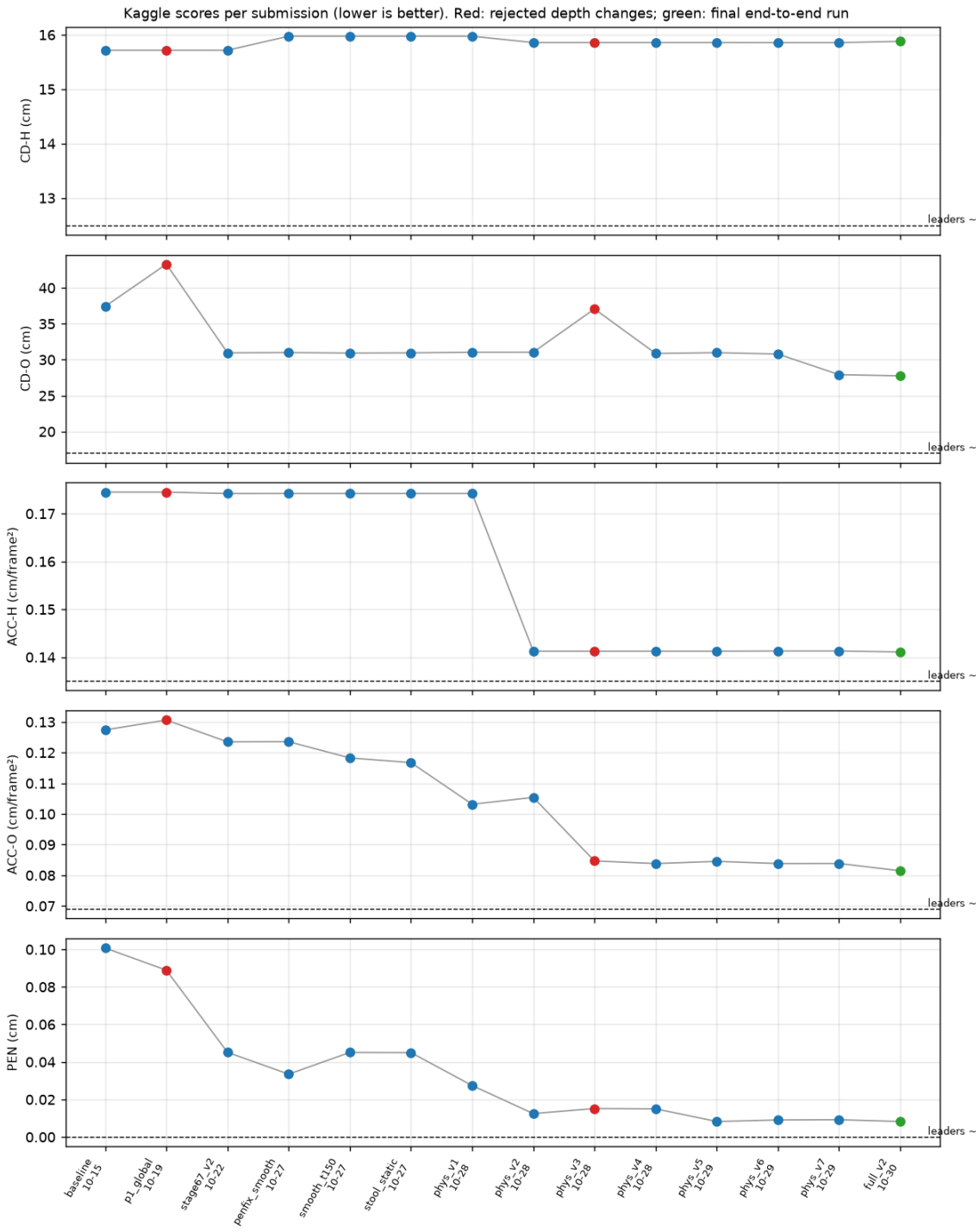


Figure 5. Kaggle scores of every submission. The CD-H step between stage67_v2 and penfix_smooth (+0.26 cm with byte-identical body rows) is attributed to a scorer-side change; CD-H values are only compared within the same period.

#	Submission	Change (one per submission where possible)	CD-H	CD-O	ACC-H	ACC-O	PEN
0	baseline	stages 1–8 as first built	15.716	37.420	0.1745	0.1274	0.1007
1	p1_global	8 episodes: object depth to the body's scale (×1.07–1.16)	15.716	**43.294**	0.1745	0.1306	0.0889
2	stage67_v2	new meshes, tracker, hidden-frame interpolation, fixed/follow placement	15.716	30.956	0.1742	0.1235	0.0451
3	penfix_smooth	object push-out on foam 15–17	15.974	30.998	0.1742	0.1236	0.0335
4	smooth_t150	λ 150 on 4 episodes	15.974	30.929	0.1742	0.1183	0.0451

#	Submission	Change (one per submission where possible)	CD-H	CD-O	ACC-H	ACC-O	PEN
5	stool_static	stools 24/26 locked (control)	15.974	30.956	0.1742	0.1167	0.0450
6	phys_v1	λ 150 everywhere + push-out where it passes held-out gates	15.974	31.043	0.1742	0.1032	0.0274
7	phys_v2	+ body smoothing σ 4 + push-out ≤ 5 cm, foam shrink	15.857	31.049	0.1413	0.1054	0.0125
8	phys_v3	+ λ 1000 + object depth ×1.04 (from the test set)	15.857	**37.062**	0.1413	0.0847	0.0152
9	phys_v4	phys_v3 without the depth change	15.857	30.871	0.1413	0.0838	0.0150
10	phys_v5	+ second push-out pass (≤ 7 cm)	15.857	30.982	0.1413	0.0845	0.0083
11	phys_v6	phys_v4 + upper-limb depenetration	15.855	30.786	0.1413	0.0838	0.0091
12	phys_v7	+ hoop floor (ep 0), hoop ellipse (ep 1), desk locks	15.855	27.918	0.1413	0.0839	0.0092
14	full_v2	fresh end-to-end run of the committed code	15.879	**27.754**	**0.1411**	**0.0815**	**0.0083**

4.1 What worked, ranked by evidence

Change	Kaggle effect	Predicted beforehand?	Evidence
Object smoothing λ 50→150→1000	ACC-O 0.117 → 0.084; CD-O –0.18	Yes (test set)	phys_v1, phys_v4
Body smoothing σ 4	ACC-H –0.033; CD-H –0.12	Yes for ACC-H (pelvis only)	phys_v2
Object push-out	PEN 0.045 → 0.015 (then 0.008)	Yes (local PEN)	phys_v1, v2, v5
Upper-limb depenetration	PEN 0.0150 → 0.0091, CD-O –0.085	PEN yes; CD-O sign no	phys_v6
Hoop floor + hoop ellipse + desk locks	CD-O –2.87	Direction yes, size no (predicted –0.3…–1.8)	phys_v7
Fresh end-to-end run	all five within pre-registered ranges	Yes	full_v2 (H14)

4.2 What failed, and why it matters

Attempt	Result	Lesson
Object depth to the body's scale (p1_global)	CD-O +5.9	Hand contact, not body scale, is the better depth cue.
Depth ×1.04 learned on the test set (phys_v3)	CD-O +6.0 while the test set predicted –4.4	Test-set placement conclusions do not transfer; our alignment there is not the scorer's (section 5.3).
Alternative models: TRELLIS.2 / Hunyuan3D-2 meshes, Video Depth Anything, CoTracker refinement	no gain over production on the gates	Shape is not the bottleneck: CD-O is 20.7 cm on the test set but 3.6 cm once the object centre is put on the reference centre.
Hoop: mask-hole repair, more/other tracker anchors, torus geometry	rejected by scored-frame gates	Better masks did not yield better 3D poses for a symmetric ring; geometry did.
Joint silhouette refinement	image fit up, reference error unchanged	The remaining lateral error is not a 2D misfit (pre-registered H5 falsified).
"Scored frames are contact frames" (stage 8 assumption)	false at the start of many spans	15 episodes start at frame 90 with the hand still reaching; contact-based steps need a contact detector.

5. How changes were validated

5.1 Isolation and gates

- Every experiment wrote to its own directory under experiments/; production outputs, earlier submissions and the official scorer were never modified. Each candidate has a manifest (inputs, commit, file hash, expected effect).
- Each step declares which arrays it may change; the merge tool rejects anything else. Every packed file was checked with the kit's validator (740,780 rows, row order equal to the sample, finite values).
- Per-episode acceptance gates (held-out PEN, mask IoU on held-out frames, self-acceleration limits, maximum shift) decide where a step is applied.
- One change per submission where possible; the experiment ledger (docs/EXPERIMENTS.md) records every submission, its prediction and its result.

5.2 Evidence classes

Evidence	Establishes	Does not establish
Unit tests (104, all passing)	math and indexing on synthetic cases; contracts such as "a rejected mesh does not drop an episode"	that a method improves the score
Local proxies (PEN, hand–object gap, contact share, self-acceleration)	physical consistency; Kaggle PEN was 81–93 % of the local value	CD or ACC error against the hidden reference (the gap proxy agreed with Kaggle on phys_v3; the test set did not)
Human review sheets	visible failures: wrong hands, hidden-depth cheats, wrong contact interpretation	metric size
Local test set (Track 2)	smoothing strength (confirmed twice)	object placement (contradicted by Kaggle)
Kaggle	hidden-reference effect of the whole file	per-episode attribution

5.3 Local test set and its limits

Track 2 Tier 1 contains 30 other episodes from the same capture setup with the multi-view reference. We ran the unchanged pipeline on six of them in a separate directory that contained only videos, metadata and our chosen scored frames; reference data lived in a second directory read only by the evaluator, and a test checks that no reference file is reachable from the pipeline's directory. Because Track 2 has no camera extrinsics and only a root pose for the human, our camera was registered to the reference world through the pelvis trajectory. That alignment is not the scorer's first-frame whole-body alignment, and episodes where the person barely moves cannot be registered (one of six was excluded). Smoothing conclusions transferred to Kaggle; the object-depth conclusion did not, and a first-frame emulation (pre-registered H1) was inconclusive. Test-set placement results are therefore not used.

Rule compliance. The test set was used for evaluation and for choosing λ and the body σ; the team stated that Track 2 data may not be used in any submission. Whether evaluation-only use is acceptable must be confirmed with the organisers. If it is not, the affected hyperparameters should be re-justified from Track 1 evidence alone (λ and σ were also confirmed directly on the Track 1 leaderboard).

5.4 Pre-registered predictions

From 29 October, predictions were written into the ledger before each upload (ranges per metric and a failure condition). Outcomes: phys_v6 inside the ranges for four metrics, CD-O better than predicted; phys_v7 CD-O far better than predicted (–2.87 vs –0.3…–1.8); full_v2 inside all five ranges; the joint-fit hypothesis J was not submitted because its test-set check failed.

6. Human review

Review sheets (camera view, top-down view, first scored frame of all episodes, contact-failure episodes) caught problems no metric did: hands bent out of objects, a hand hidden behind a desk, the hoop lying at the feet in episode 0, the held hoop in episode 1, and which objects are static. Team-supplied facts used in the pipeline: stools 24/26 static; desk static in the first part of episode 10 and until the end of episode 11; carts move; hoop diameter about 67–72 cm (used only in a rejected variant); episode-0 late masks lack the hole.



Figure 6. First scored frame of every episode (the frame the scorer aligns on), episodes 0–29 row by row, from the phys_v6 review sheet. Body blue, hands red, object orange.

7. Reproducibility

7.1 The fresh run

The committed code was run on all 30 videos in a new directory (/local-data/ivan/v2d_full), using only the released videos, model weights and the submission kit. Stages 1–2 took about 6 h, stages 3–8 about 13 h, post-processing about 1 h. The run was not uninterrupted: it exposed four bugs, each fixed with a test, after which the affected stages were re-run:

#	Symptom	Cause	Fix	Nature
1	only 28 of 30 episodes reached the submission while every stage reported 0 failures	a mesh rejected by stage-5e validation was never stamped, so stages 6–8 silently skipped the episode	keep the mesh and stamp it with a WARN (STRICT_5E=1 restores the stop)	completion fallback, not a quality fix
2	episode 1 dropped in stage 6	no anchor reached IoU 0.4 with the poorer mesh	keep the best anchor with a WARN (STRICT_6=1)	completion fallback
3	hoop episodes far worse (hand gap 64 → 220 cm)	stage 5e chose a different hoop mesh than production, from a frame inside the stretch where SAM2 fills the hoop's hole	reviewed override (frame-300 candidate)	dataset-specific choice
4	the override was ignored	the 5e stamp did not include object_overrides.json	add it to the stamp inputs	genuine cache bug

With the fixes, full_v2 reproduced the earlier best result on Kaggle (CD-O 27.75 vs 27.92). The models are not bitwise deterministic, so every episode differs slightly from production; the hoop floor factor was re-derived automatically (×1.197 vs ×1.193).

7.2 Remaining gaps before handing the code to someone else

- The top-level launcher for the full run lives outside the repository and does not stop on every failure exit code; a committed, fail-fast launcher is needed.

- Completion fallbacks (bugs 1–2) keep runs going but produce warnings, not trustworthy reconstructions; a final summary should list every fallback.
- Mesh selection needs reviewed overrides for two objects; selection is not stable between runs.
- Some contract tests check source text rather than executing the failure scenario.
- Environment setup (several model-specific virtual environments, weights) is documented only in scripts.
- One repaired run is evidence of reproducibility of quality, not of determinism.

7.3 How to run

```
stages 1–8 (per stage, two GPUs):
GPU=0 EPISODES="0-29" ./run_stage.sh 1   ...   FROM=3 TO=8 GPUS="0 2" ./run_rest.sh
post-processing, packing, validation, local proxies:
/opt/venvs/sam3d_body/bin/python track1/postprocess.py --run <name> --commit <sha>
upload (kit): python3 submit/v2d_submit.py <file> --track track_1 --wait
```

8. Limitations and next steps

- **Object placement relative to the body** is the dominant remaining error (CD-O 27.8 vs 16–18 cm for the leaders). Depth placement by hand contact is biased on approach frames, and lateral errors are not visible in 2D. A contact detector, and placement driven by the body's own hands rather than by monocular depth, are the most promising directions; any change must be validated on Track 1 itself.
- **CD-H** was barely addressed (15.9 vs 12–13). Body drift over time and the first-frame alignment are untested hypotheses (H2, H9).
- **Gross single-episode errors** dominate CD-O. A systematic audit of every interaction type (feet, sitting, two-handed, held thin objects) found one (episode 0); paint roller 18/19 and stool 26 still show 10–16 cm hand gaps without an identified cause.
- **Episode-specific knowledge** (static intervals, overrides) came from human review; it is documented with reasons in the configuration but would need detectors for a fully automatic system.
- **Evaluation data use** must be confirmed (section 5.3).

Appendix A. Files and provenance

Item	Location
Final submission	experiments/full_v2/track1.parquet (sha256 d5b92f46fdb3…, commit af639e8)
Earlier best	experiments/phys_v7/track1.parquet (commit 363831d)
Experiment ledger, predictions	v2d_challange/docs/EXPERIMENTS.md
Run notes	v2d_challange/docs/STATUS_2026-10-29.md, STAGE_AUDIT_2026-10-27.md
Post-processing chain	v2d_challange/track1/postprocess.py, postprocess_config.json
Fresh-run directory and logs	/local-data/ivan/v2d_full (logs/driver*.txt, experiments/full_v2/postprocess.log)
Review sheets	experiments/review/, experiments/phys_v6/hd_*/plots/, experiments/hoop_conic_ep1_img/
Local test set (evaluation only)	/local-data/ivan/v2d_t2 (pipeline inputs), /local-data/ivan/v2d_t2eval (reference, evaluator outputs)
Code repository	github.com/ivanvoid/v2d_challange, branch beta-models (HEAD 8d0c7ef at writing)
This report	report/ (make_figures.py, build_report.py, figures/)

Appendix B. Main tools written during this work package

Tool (track1/)	Purpose
ablate_pose_scale.py, stage8_scale.py --pin	re-smooth objects while keeping stage-8 depth placement
smooth_body.py	body smoothing that is safe across Euler-angle representation jumps
pen_squeeze.py, assemble_phys.py	object push-out and per-episode acceptance
hand_depen.py	upper-limb depenetration with image and motion constraints, review sheets
stationary_stool.py, static_lock.py	single-pose locks for static (intervals of) objects
hoop_conic.py, hoop_floor.py	hoop pose from the image ellipse; hoop at sole level when lying flat
object_depth_scale.py	scale an object about the camera (used in the failed depth experiment)
joint_fit.py	differentiable silhouette refinement (disabled)
merge_episodes.py, postprocess.py	assembly with provenance; the reproducible post-processing chain
t2_testset.py, t2_diag.py	local test set: setup, evaluation, overlays, alignment and depth diagnostics
review_sheets.py	first-frame and contact review sheets

Appendix C. Acknowledged prior work

Ideas were taken from HORIA (interaction-aware 4D reconstruction; phase-gated contact and collision losses), RHINO (closest-5 % hand contact and SDF penetration losses), Open4DHOI (stationary-object anchoring), the RHOBIN challenge report and PhysDynPose; no code from these projects is included in the submission. CARI4D weights were not available to us; its reported PEN (0.00423) is the only baseline number we compare against.